Introduction

EM1421 - Social media and web analytics

{Slides courtesy of Prof. Fabiana Zollo}

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Ca' Foscari University of Venice



About Me

- I am an Associate Professor of Computer Science in the Dept. of Environmental Sciences, Informatics and Statistics.
- Ca' Foscari web page: https://www.unive.it/data/persone/29529719
- Personal web page: https://pierlui92.github.io/
- My research interests: Artificial Intelligence, Computer Vision

About This Class

- Web page
- 30 hours of frontal lessons
- Class Hours:
  - MON: 15:45 - 17:15, ROOM 2B, San Giobbe
  - TUE: 12:15 - 13:45, ROOM 2B, San Giobbe
  - WED: 12:15 - 13:45, ROOM 2B, San Giobbe
- The course material will be available on moodle

Office hours

- **When**: Upon request by filling the Google form available on Moodle
- **Where**: Online (Preferred). In person at Scientific Campus



Main Referral Texts

- Networks, crowds, and markets. David Easley and Jon Kleinberg, Cambridge University Press, 2012
- The complete pre-publication draft of the book is available for free here
- Network Science. Albert-Laszlo Barabasi, Cambridge University Press, 2016
- Papers and other resources reported in class

Course Requirements

- **Oral exam**
- Theoretical questions and exercises on topics covered during the course (30%)
- Discussion of the project previously agreed upon with the instructor (70%)



About the Project

- Detailed guidelines & instructions are available on Moodle.
- Let's review them together!



Any Questions?



About this Class

- Social media
- Graph Theory
- Social Network Analysis
- Information Networks
- Topic extraction
- Sentiment Analysis



Introduction

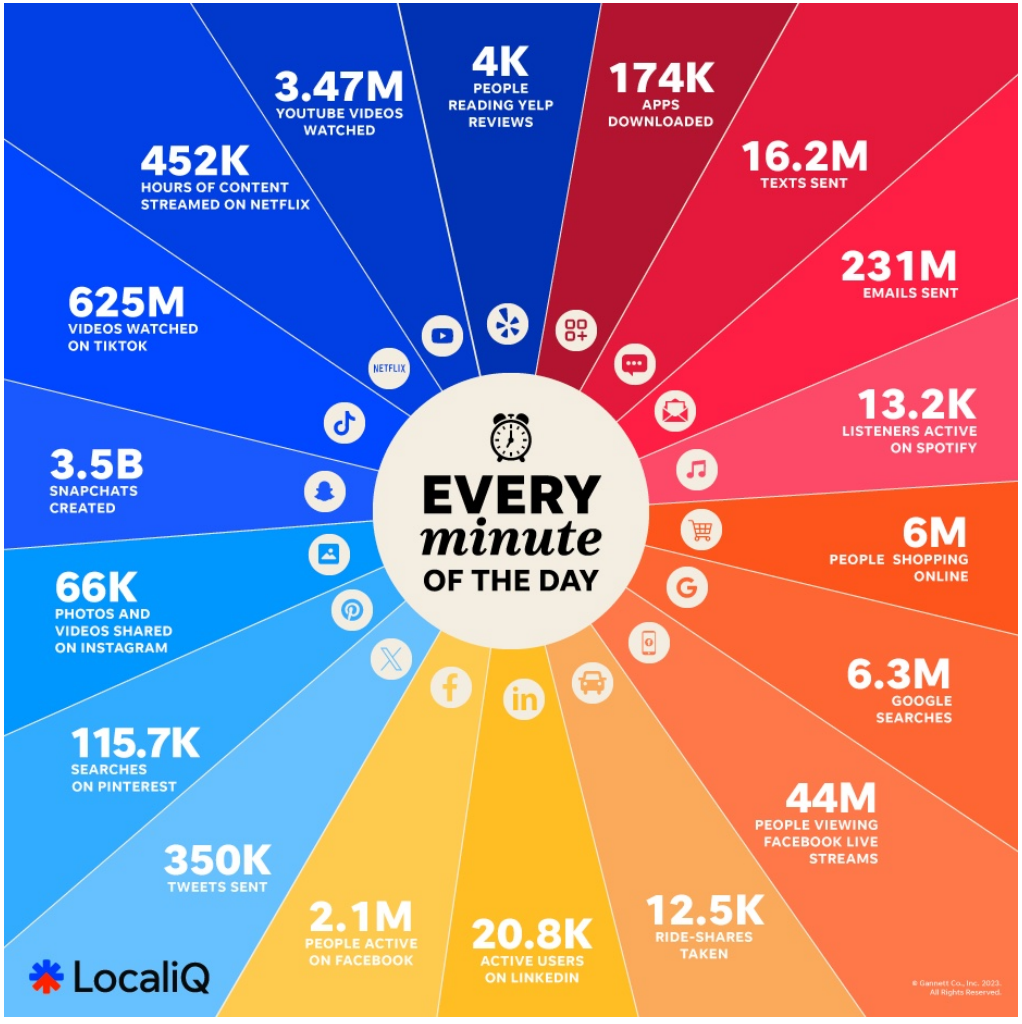
- This course is about **social media** and web analytics
- Understanding the **digital environment** is crucial before starting any kind of analysis..



Social Media and the Web



A New Information Space



A Shift of Paradigm



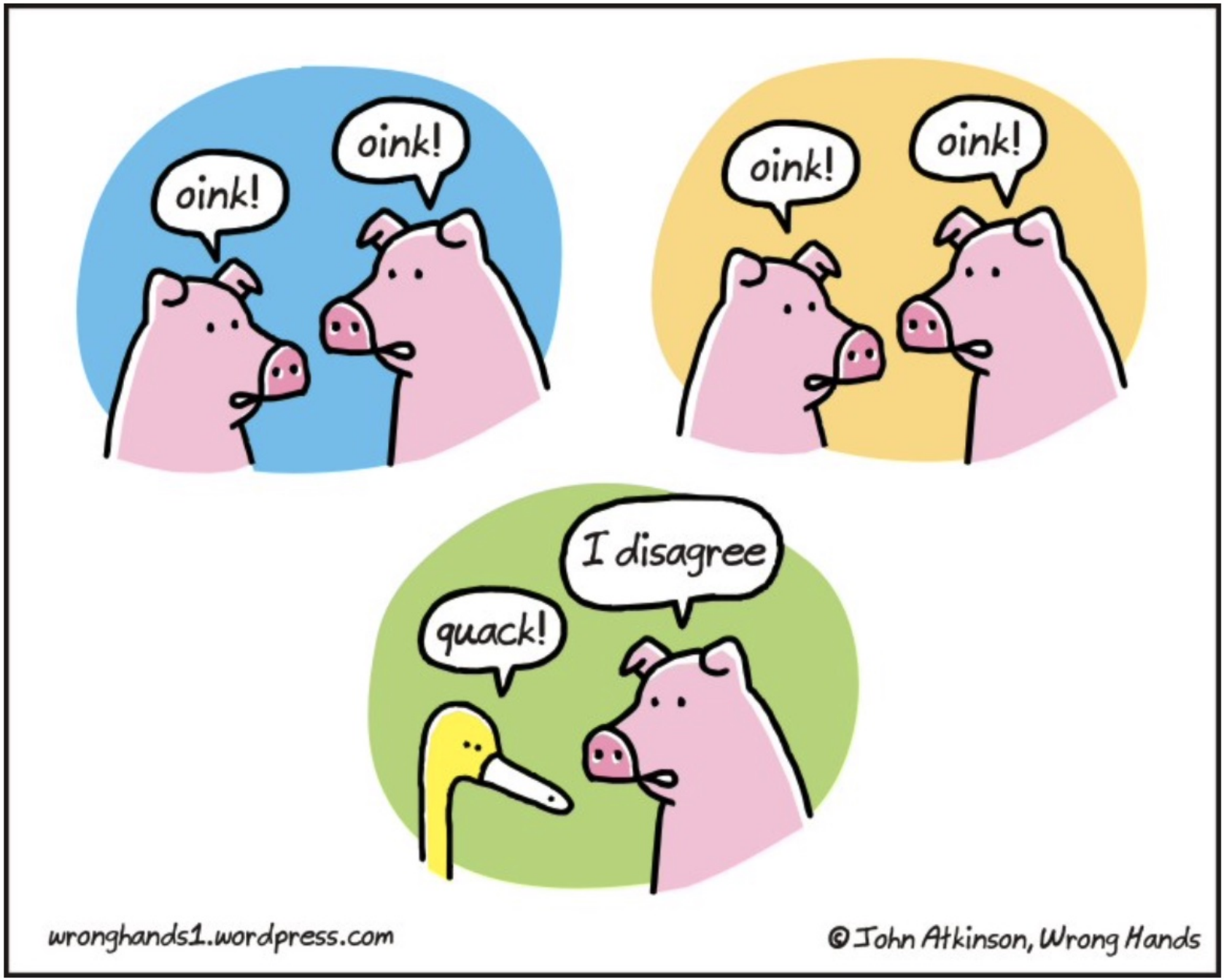
Traditional Media

New Media

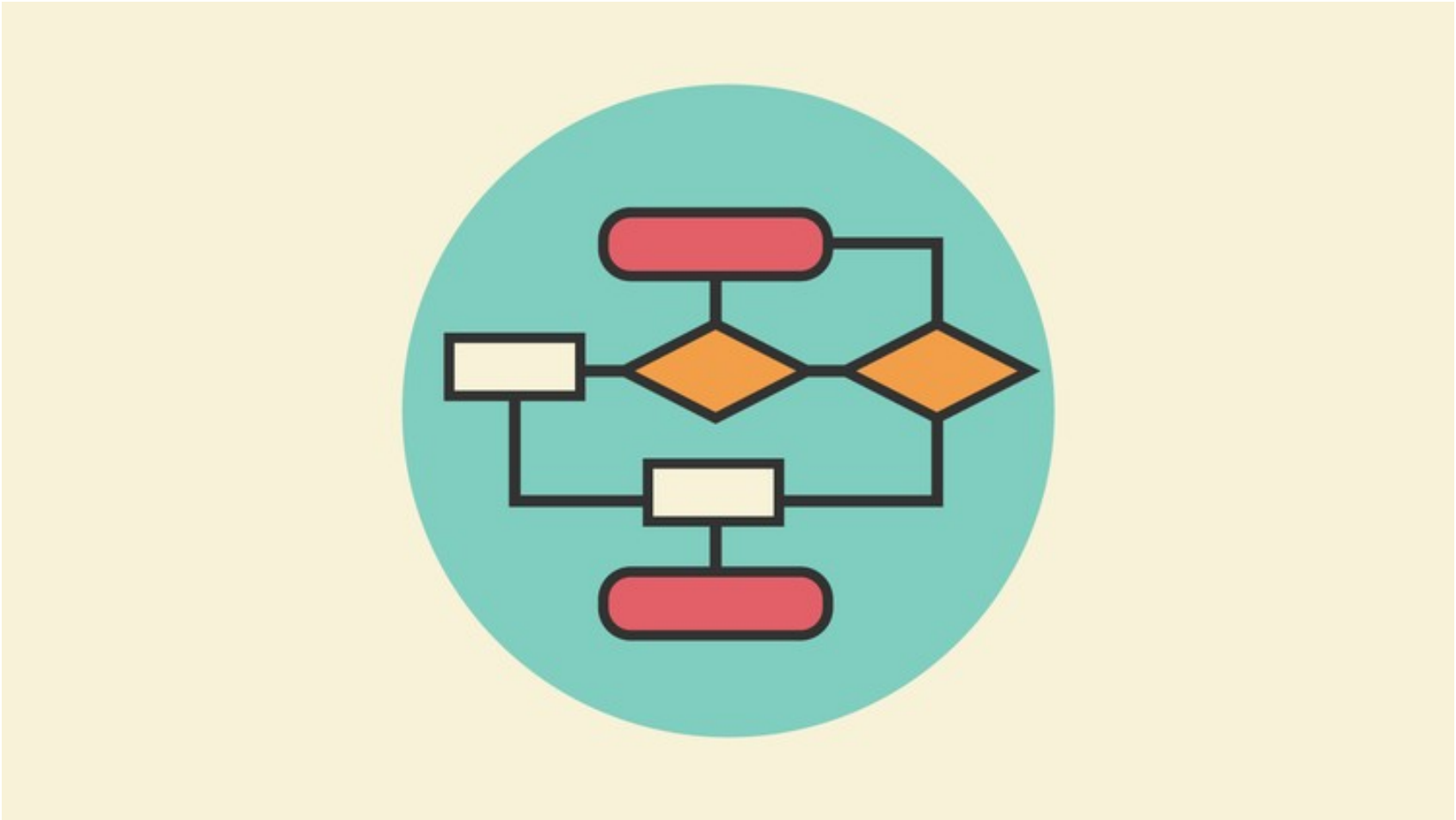
Information Processing



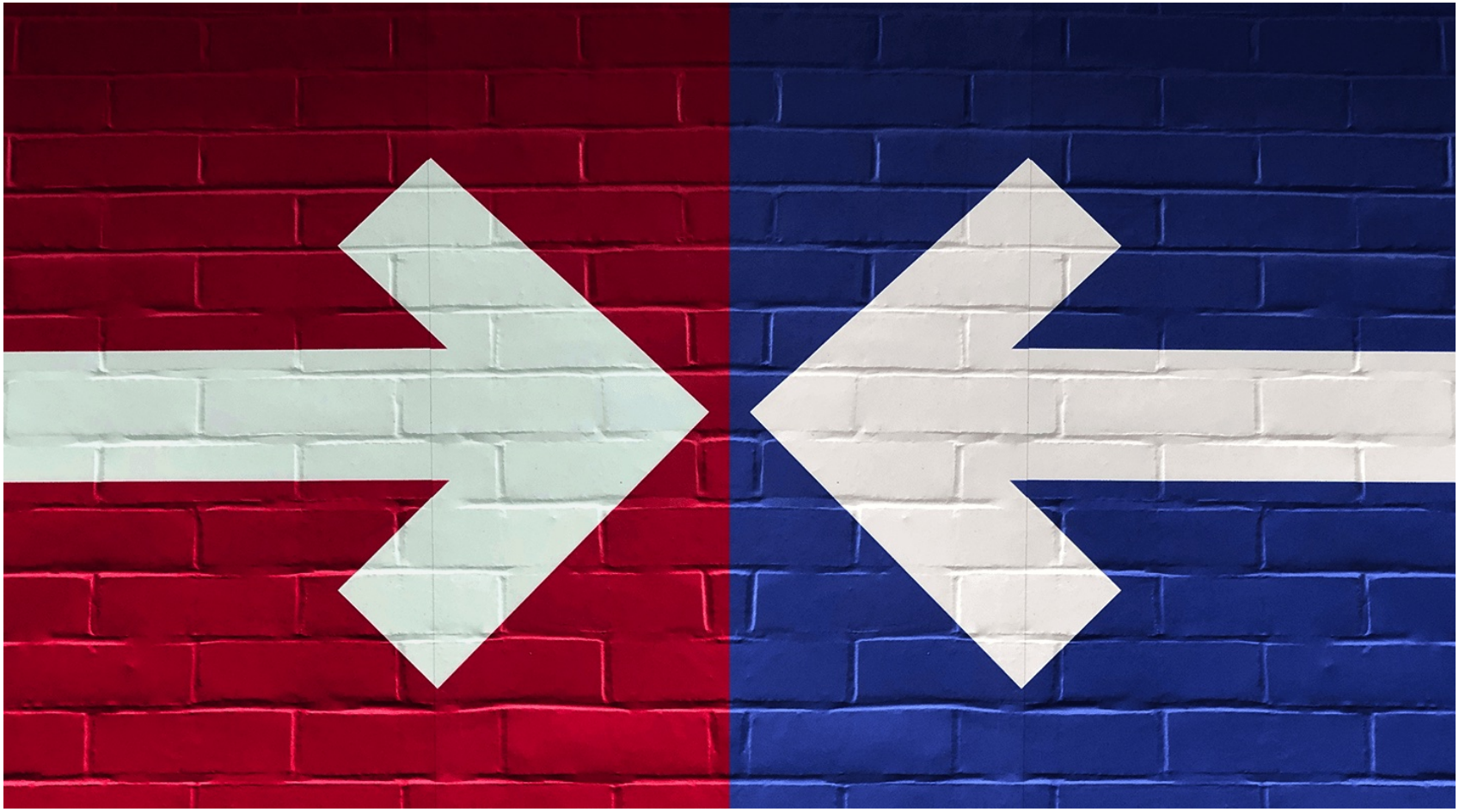
Biases


oink!
oink!
oink!
oink!
quack!
I disagree
wronghands1.wordpress.com
©John Atkinson, Wrong Hands

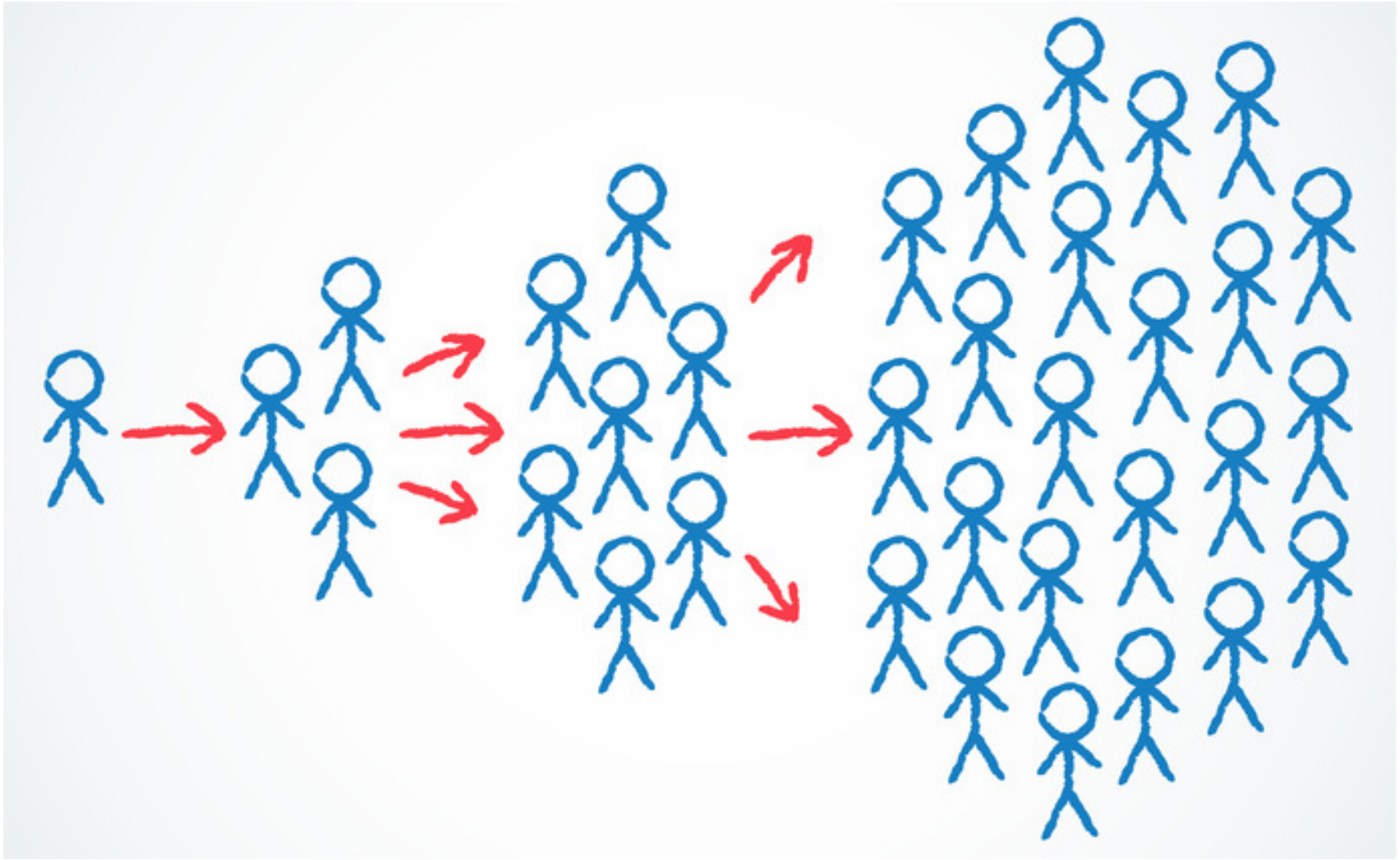
Algorithms



Polarization

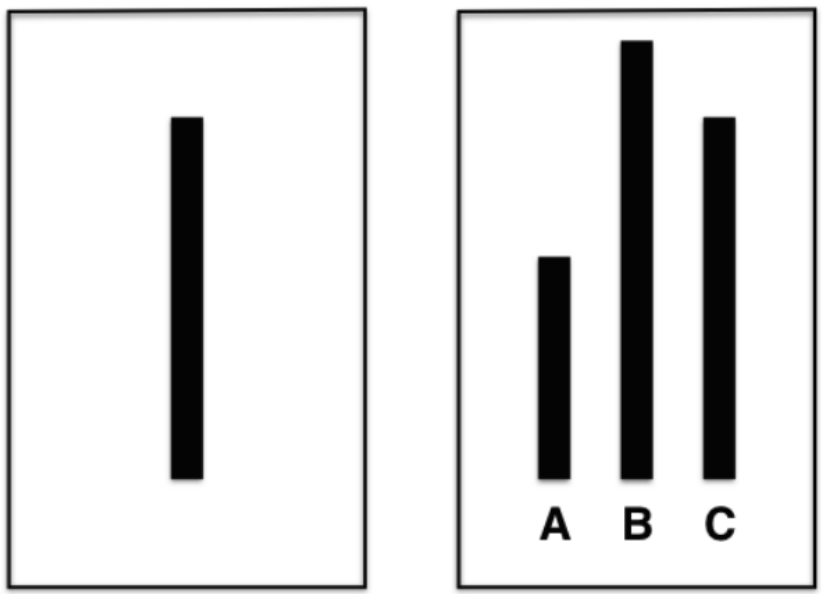


Social Contagion

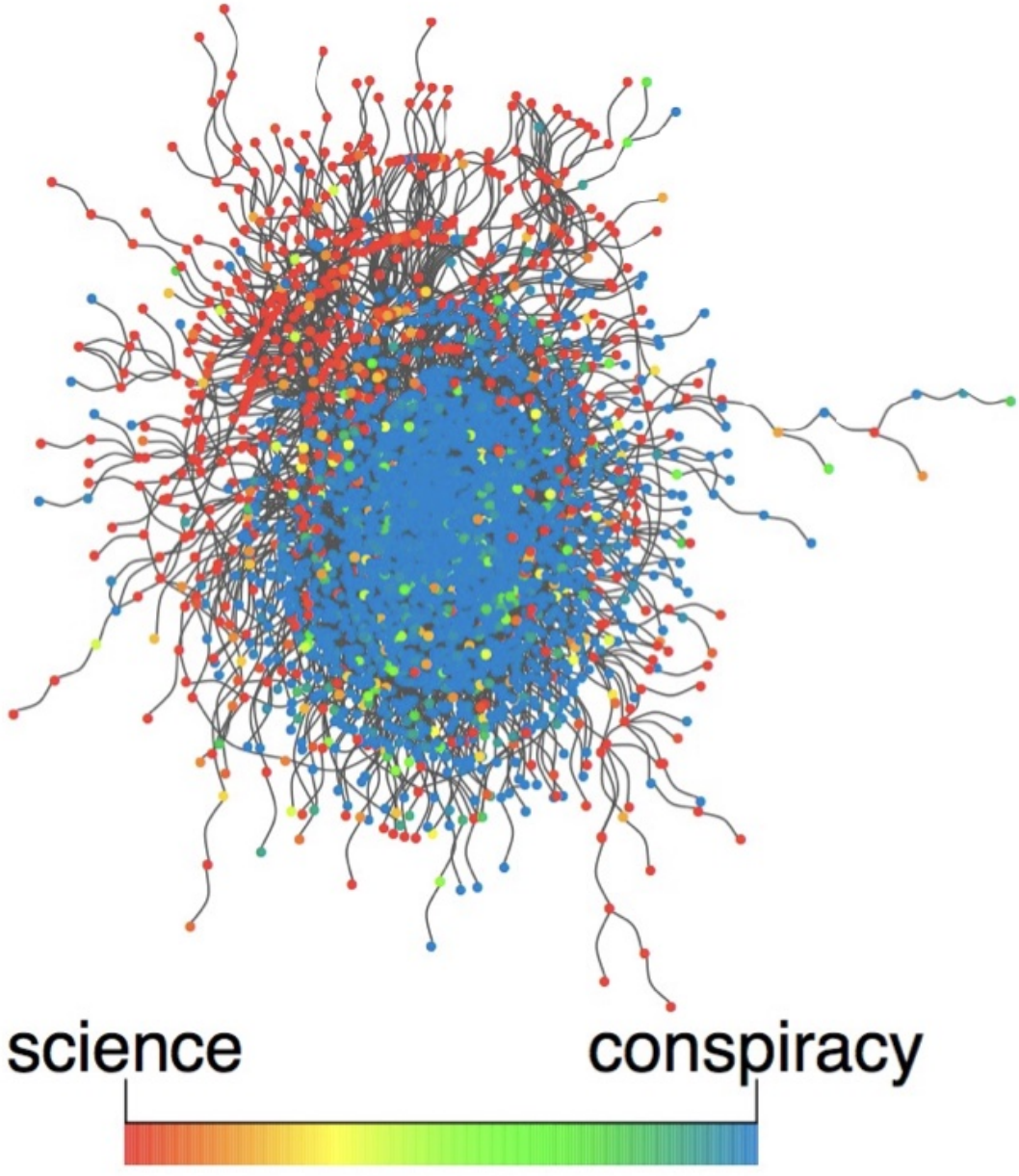


Group Polarization

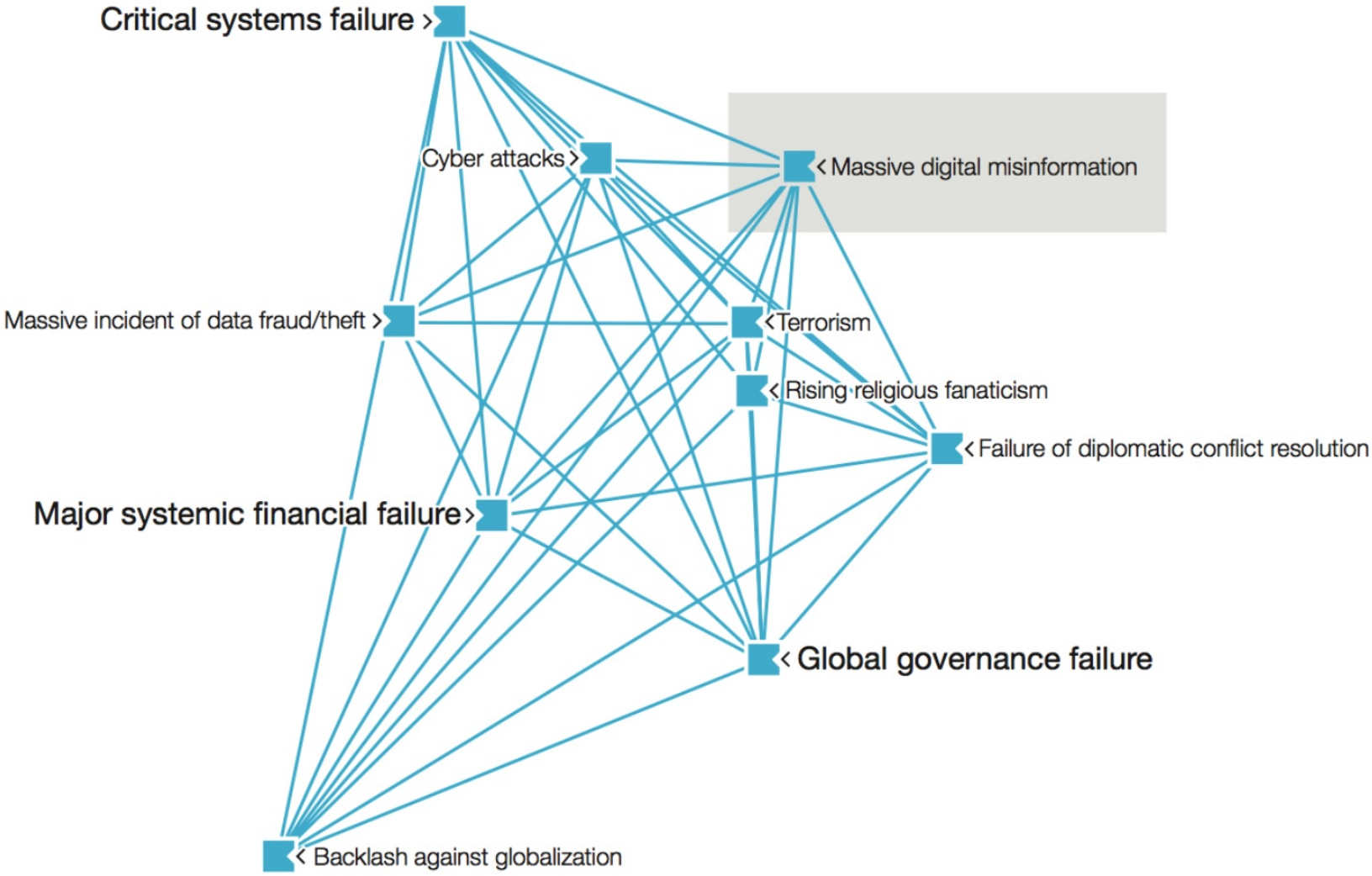
- When like-minded people discuss, they reinforce their shared views thus ending up in a more extreme position (Sunstein, 1999)
- Asch Experiment, 1955



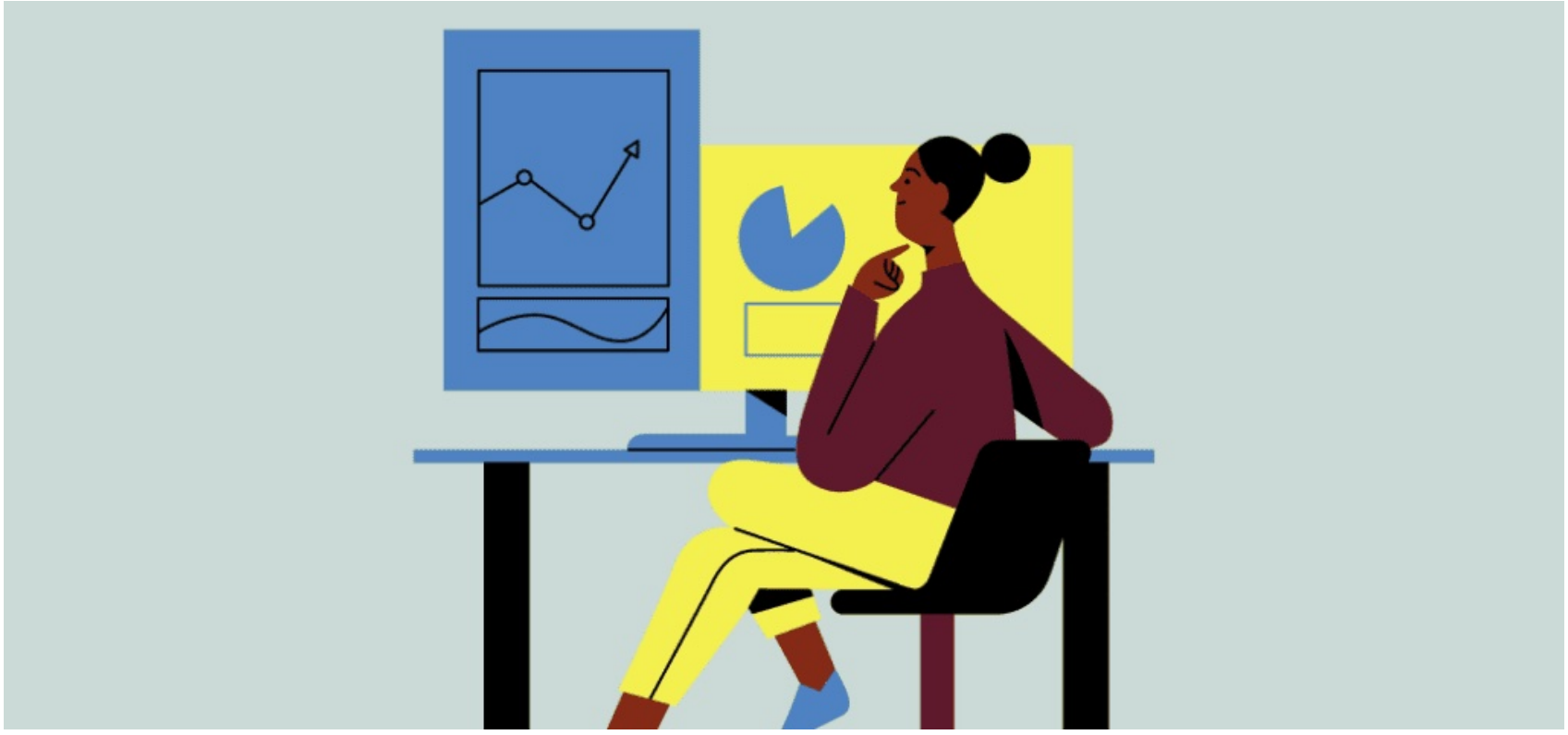
Echo Chambers



Massive Digital Misinformation



How Can we Analyze These Dynamics?

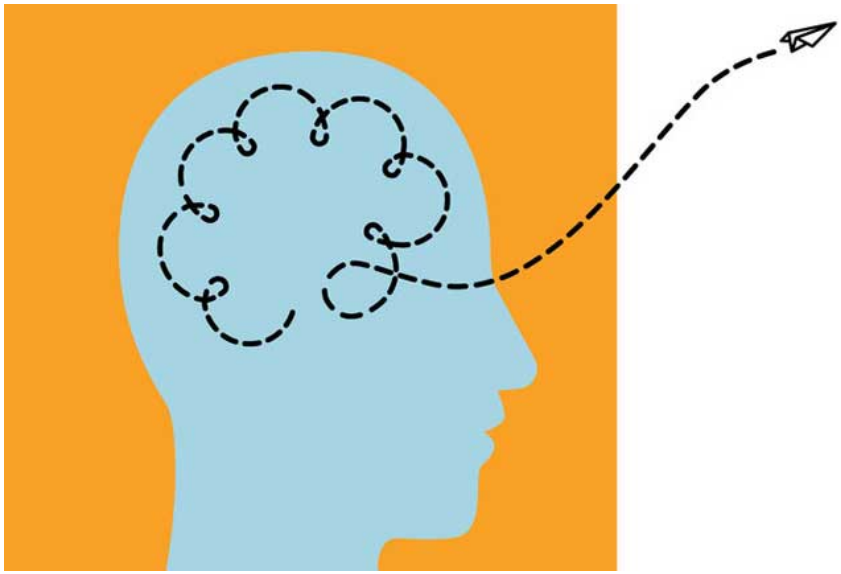


Quantitative Analysis of Social Phenomena

- Social media and our digital lives contribute to the generation of huge amount of data
- This data can be analyzed
  - To look for hidden patterns, trends or other insights
  - To investigate the **behavior** of individuals and groups of individuals

Social Networks

- In particular, we can pose questions, formulate, and validate theories about **social networks**
- What's the first thing that comes to your mind when I say *social network*?



Social Networks

- Let's keep aside the term *social* for now..
- What is a **network**?



What is a Network?

- Networks form a very generic **data model** with extensive applications to a variety of fields

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- Networks form a very generic **data model** with extensive applications to a variety of fields
- A network consists of a set of **objects** and a collection of **links** identifying some relationship between these pairs of objects
- The assumption is that a **connection** exists between these objects that will help us better understand what we are studying

What is a Network?

- In mathematics, networks are called **graphs**, the set of objects are known as **vertices** or **nodes**, and the links between them are referred to as **edges**
- The study of these objects constitute a sub-field of mathematics known as **graph theory**

Basic Notation of a Graph

- $G = (V, E)$
  - V: nodes or vertices
  - E: links or edges
- $|V| = N$ size of graph
- $|E| = L$ number of links

Network Science

A definition from Wikipedia:

> ***Network science*** *is an academic field which studies* ***complex networks*** *such as telecommunication networks, computer networks, biological networks, cognitive and semantic networks, and* ***social networks****, considering distinct elements or actors represented by* ***nodes*** *and the connections between the elements or actors as* ***links****.*

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Characteristics of Network Science

- **Interdisciplinary:**address problems from disciplines other than computer science
- **Empirical** and data-driven: based on the observation of networks
- Quantitative, mathematical, **computational**

Social Networks



Social Networks

- The collections of social ties among individuals have grown steadily over the course of human history
- Technological advances facilitated distant travels, global communication, digital interaction
- Networks can be used to study these interactions
  - Nodes are individuals and links denote personal relationships

Social Network Analysis - The first example

- **Moreno's sociograms** are frequently considered as the first examples of social network analysis and visualization
- Mapping the social affinities of a group of individuals, Moreno's first sociograms visualize the relationships between pupils in a classroom

Moreno's Sociograms

Who wants to be sitting next to whom?

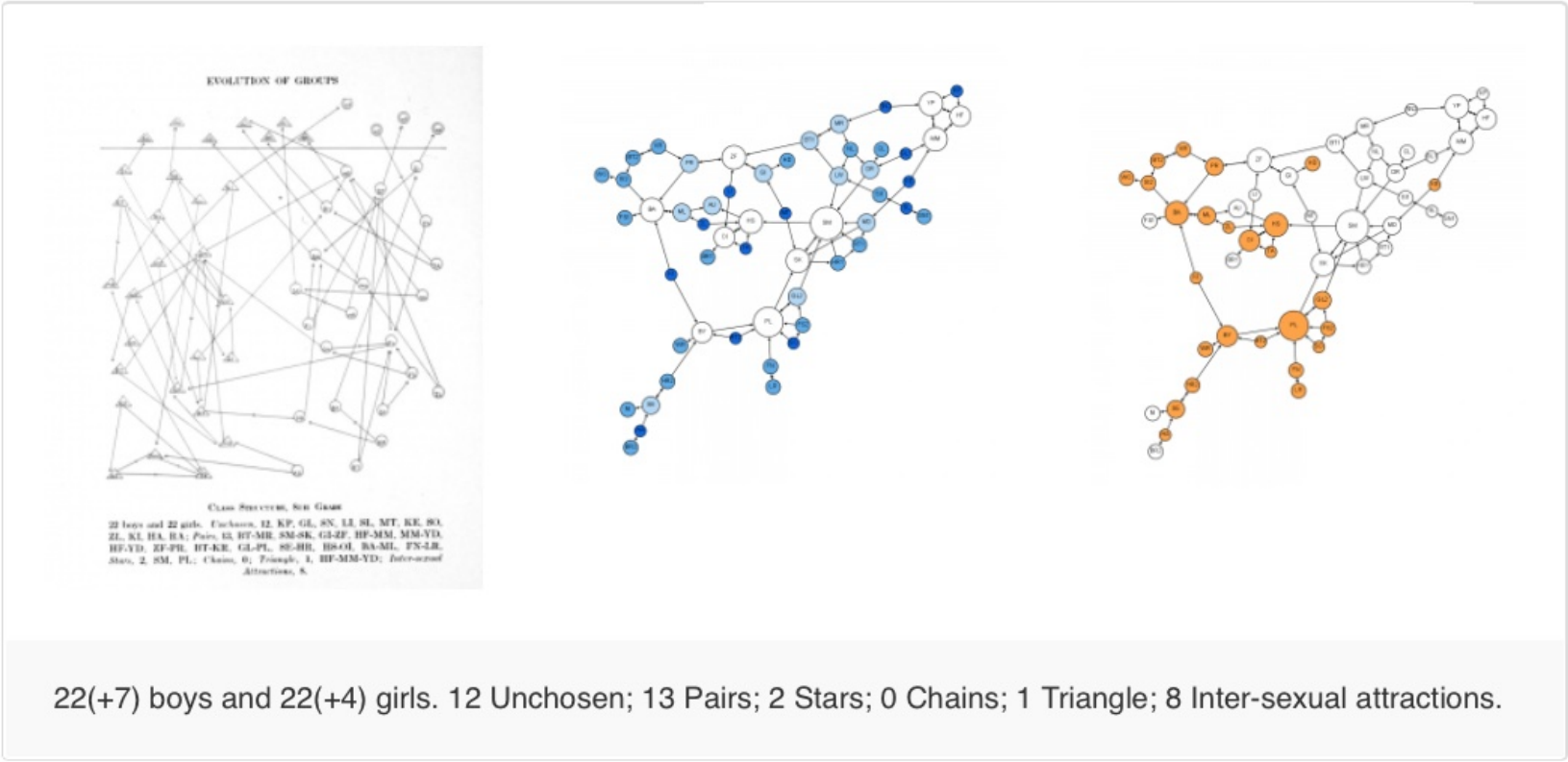
Each child can choose two others, for results that suggest that sociabilities are changing over time

- The proportion of attractions between boys and girls decreases
- Community structures are formed and then disappear, etc.

Moreno, J. L. (1934). Who shall survive?: A new approach to the problem of human interrelations.

Moreno's Sociograms

Class Structure, 8th Grade



Left: original network by Moreno. **Center:** redesigned network, indicating the number of incoming connections (dark blue=0, white=3 or more). **Right:** redesigned network, highlighting girls (orange) and boys (white). *Source*

Zachary's Karate Club

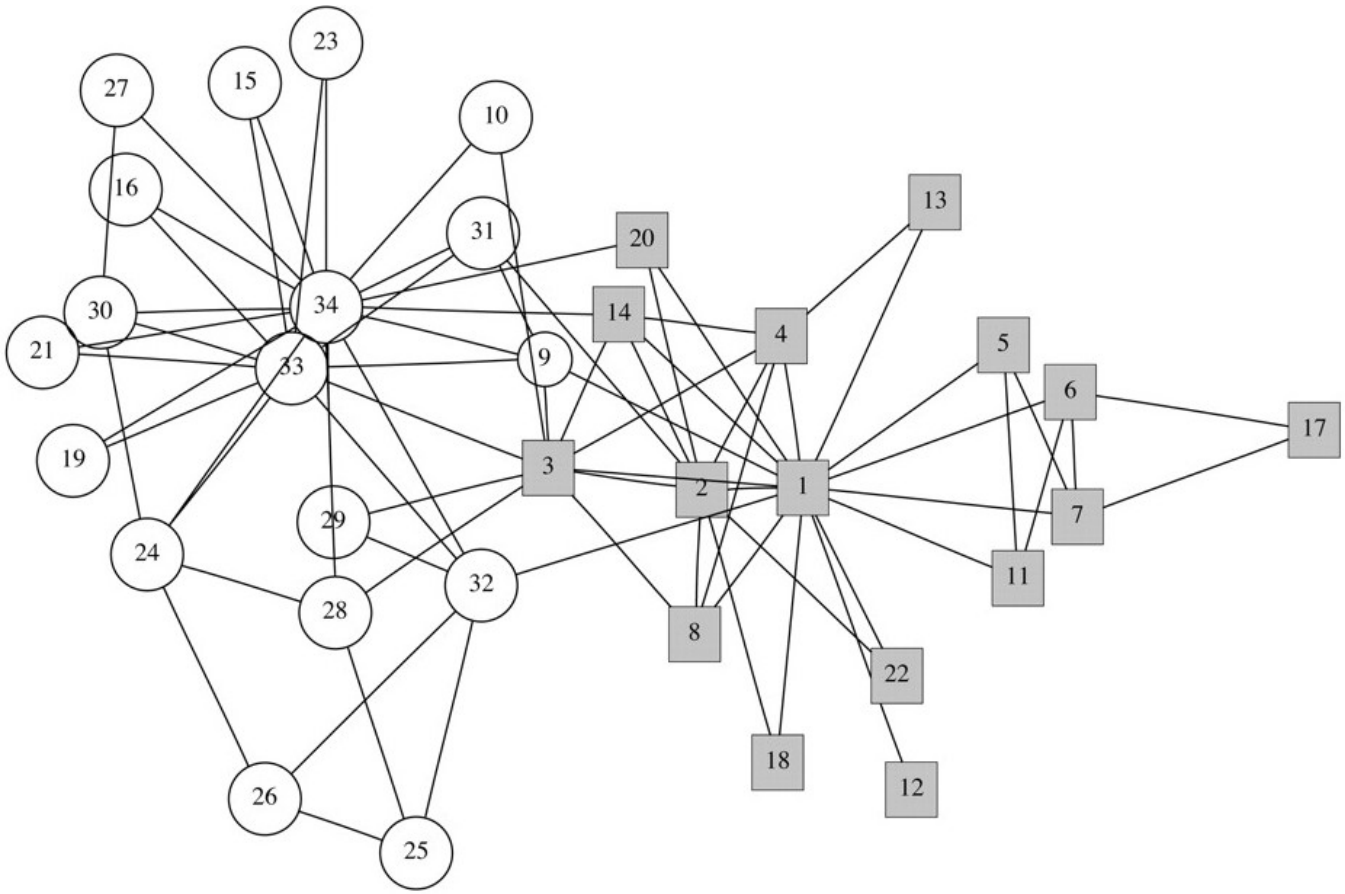
- Zachary's karate club is a social network of a university karate club described in the paper *Zachary W. (1977). An information flow model for conflict and fission in small groups. Journal of Anthropological Research, 33, 452-473*
- Example of how social networks can capture the sources of conflict within a group

Zachary's Karate Club

- Zachary observed 34 members of a karate club over a period of 2 years
- During the course of the study, a disagreement developed between the administrator of the club and the club's instructor
- In the end, the instructor left and started a new club, taking about a half of the original club's members with him
- Zachary constructed a network of friendships between members of the club, using a variety of measures to estimate the strength of ties between individuals

Zachary's Karate Club

- The instructor (node 1) and the administrator (node 34) are not friends with each other
- Most people are only friends with one or the other (noninteracting clusters)

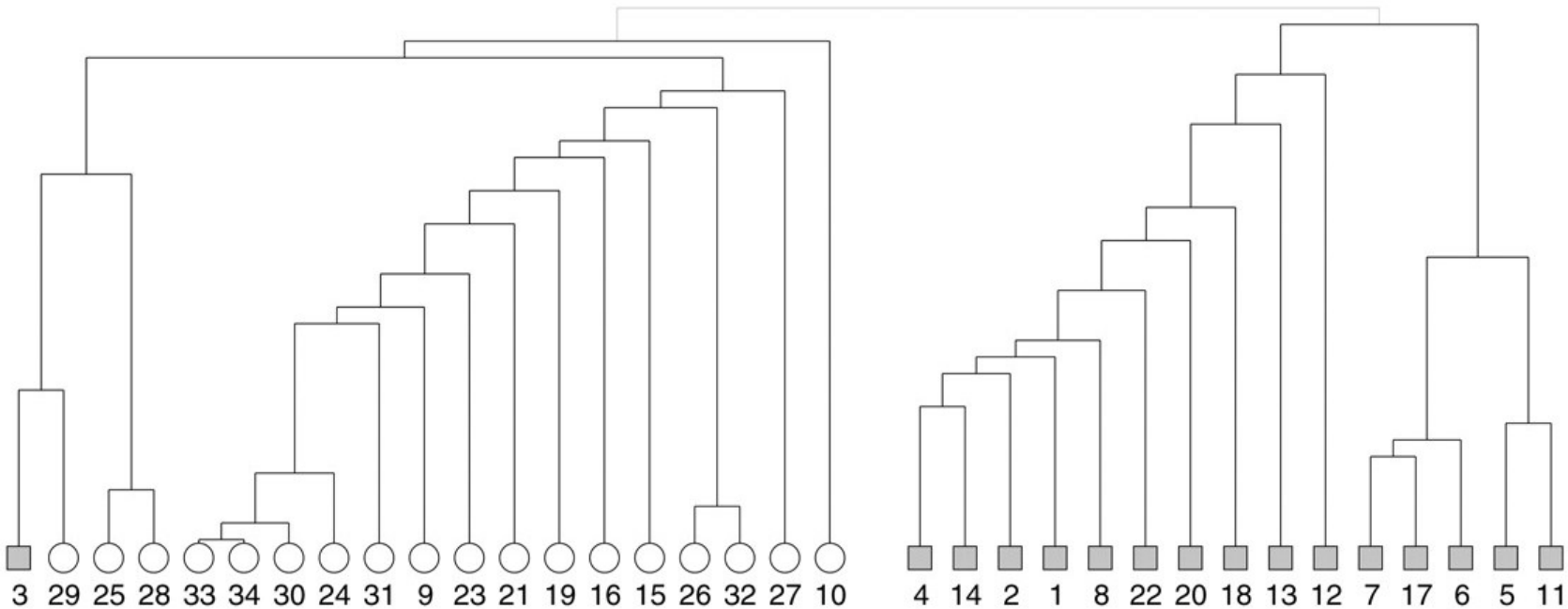


Zachary's Karate Club

- The network became a popular example of community structure in networks after its use in *Girvan M. & Newman M.E.J. (2002). Community structure in social and biological networks. PNAS. 99:7821-7826*
- In their article, Girvan & Newman highlight a property that is found in many networks, the property of community structure, in which network nodes are joined together in tightly knit groups, between which there are only looser connections, and propose a method for detecting such communities

Zachary's Karate Club

The Figure below shows the hierarchical tree of communities produced by Girvan & Newman's method

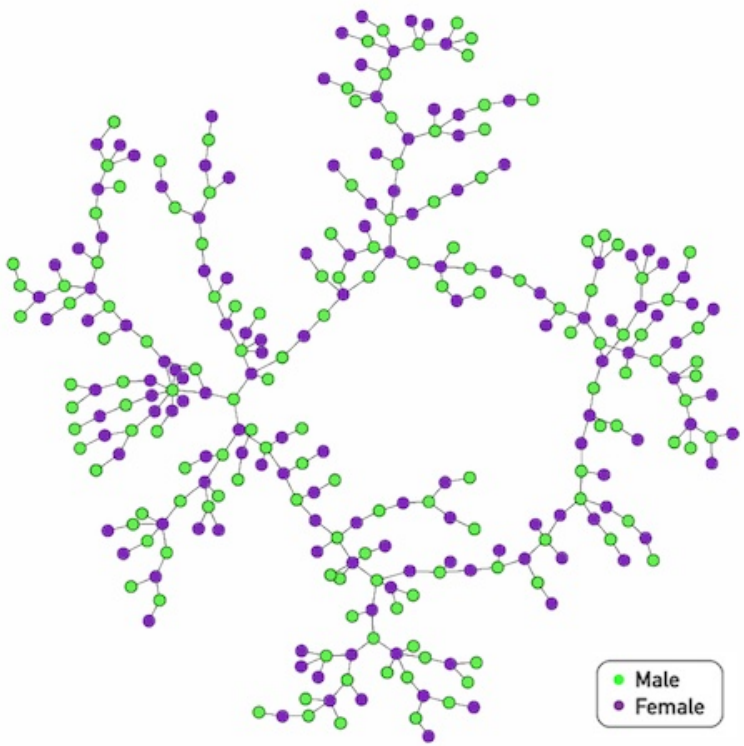


Zachary's Karate Club

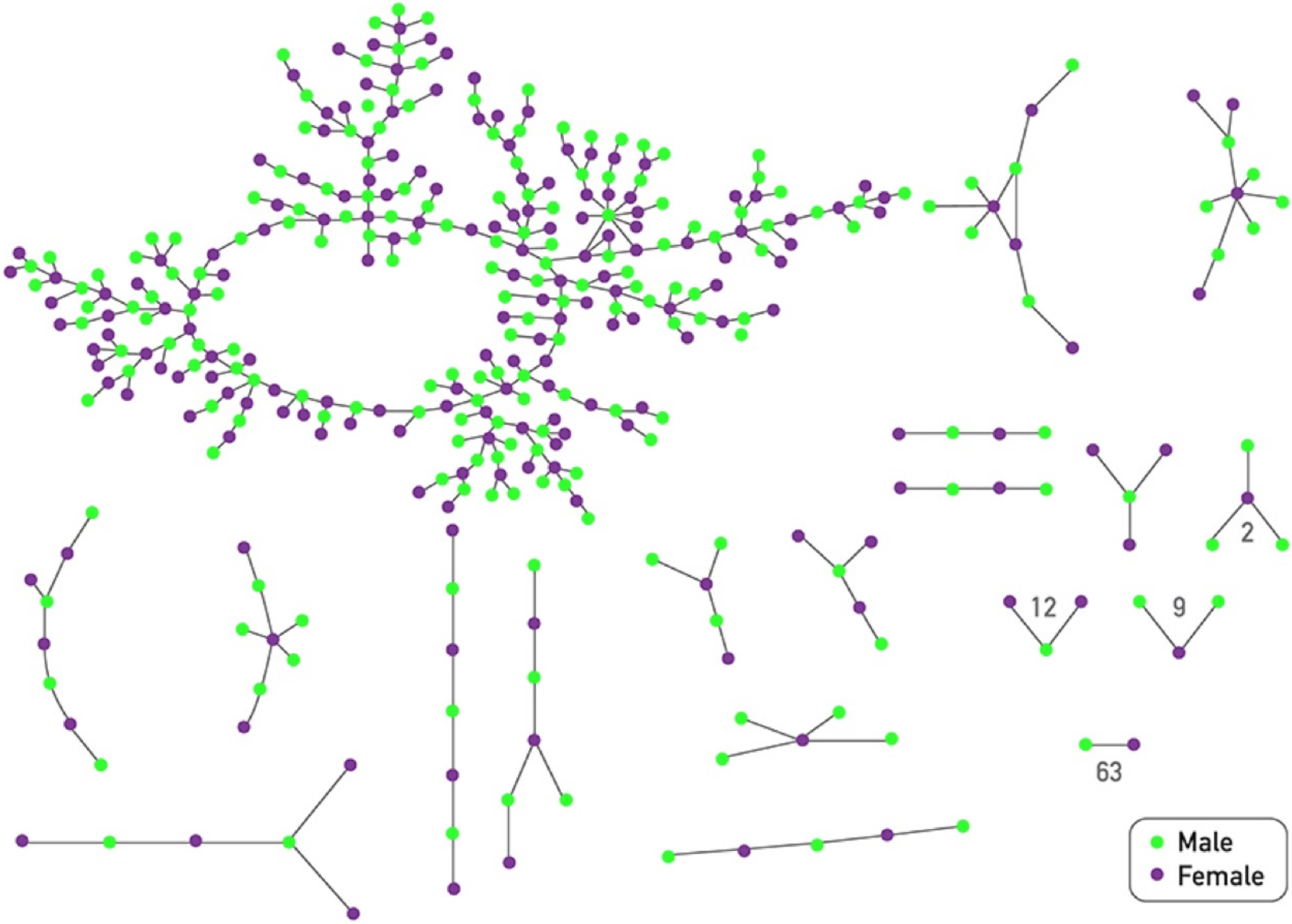
- The most fundamental split in the network is the first one at the top of the tree, which divides the network into two groups of roughly equal size
- This split corresponds almost perfectly with the actual division of the club members after the break-up
- Only one node, node 3, is classified incorrectly
- In other words, the application of the algorithm to the empirically observed network of friendships is a **good predictor** of the subsequent **social evolution** of the group

Chains of Affection

- Bearman, P. S., Moody, J., & Stovel, K. (2004). Chains of affection: The structure of adolescent romantic and sexual networks. American journal of sociology, 110(1), 44-91
  - Describe the structure of adolescent romantic and sexual networks in an American high school (in the past 18 months)
  - Accounts for the emergence of this structure
  - Links the observed structure to the diffusion dynamics of disease



Chains of Affection



Computational Social Science

- Sociology has developed fundamental insights into the structure of social networks..
- ..but its network methodology was refined in the domains and scales where data collection was possible (lab experiments, focus-groups, surveys)
- The availability of (digital) data today offers great opportunities to pose/answer questions and formulate/validate theories about social networks

Computational Social Science

Lazer et al. (2009) Computational Social Science, Science, 323:721-723

SOCIAL SCIENCE

Computational Social Science

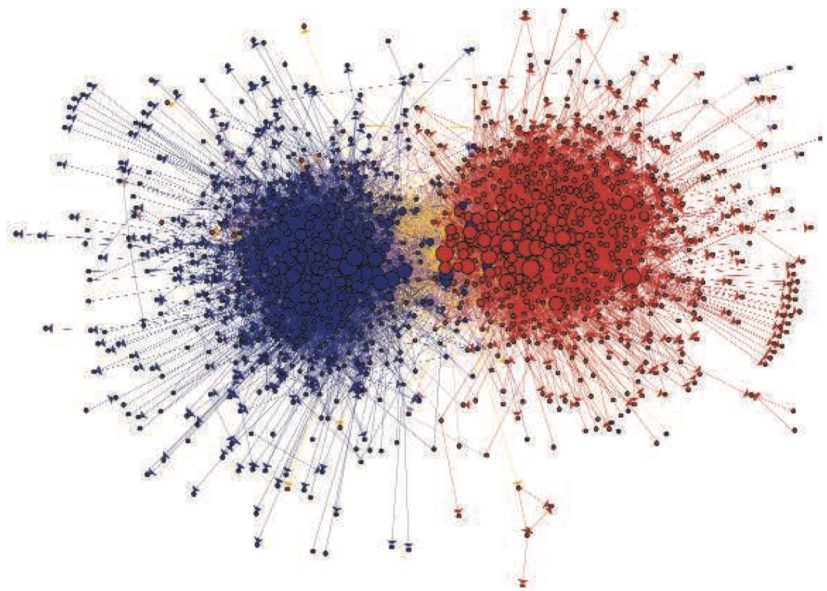
David Lazer,[1] Alex Pentland,[2] Lada Adamic,[3] Sinan Aral,[2,4] Albert-László Barabási,[5] Devon Brewer,[6] Nicholas Christakis,[1] Noshir Contractor,[7] James Fowler,[8] Myron Gutmann,[3] Tony Jebara,[9] Gary King,[1] Michael Macy,[10] Deb Roy,[2] Marshall Van Alstyne[2,11]

A field is emerging that leverages the capacity to collect and analyze data at a scale that may reveal patterns of individual and group behaviors.

We live life in the network. We check our e-mails regularly, make mobile phone calls from almost any location, swipe transit cards to use public transportation, and make purchases with credit cards. Our movements in public places may be captured by video cameras, and our medical records stored as digital files. We may post blog entries accessible to anyone, or maintain friendships through online social networks. Each of these transactions leaves digital traces that can be compiled into comprehensive pictures of both individual and group behavior, with the potential to transform our understanding of our lives, organizations, and societies.

The capacity to collect and analyze massive amounts of data has transformed such fields as biology and physics. But the emergence of a data-driven "computational social science" has been much slower. Leading journals in economics, sociology, and political science show little evidence of this field. But computational social science is occurring—in Internet companies such as Google and Yahoo, and in government agencies such as the U.S. National Security Agency. Computational social science could become the exclusive domain of private companies and government agencies. Alternatively, there might emerge a privileged set of academic researchers presiding over private data from which they produce papers that cannot be critiqued or replicated. Neither scenario will serve the long-term public interest of accumulating, verifying, and disseminating knowledge.

What value might a computational social science—based in an open academic environment—offer society, by enhancing understanding of individuals and collectives? What are the



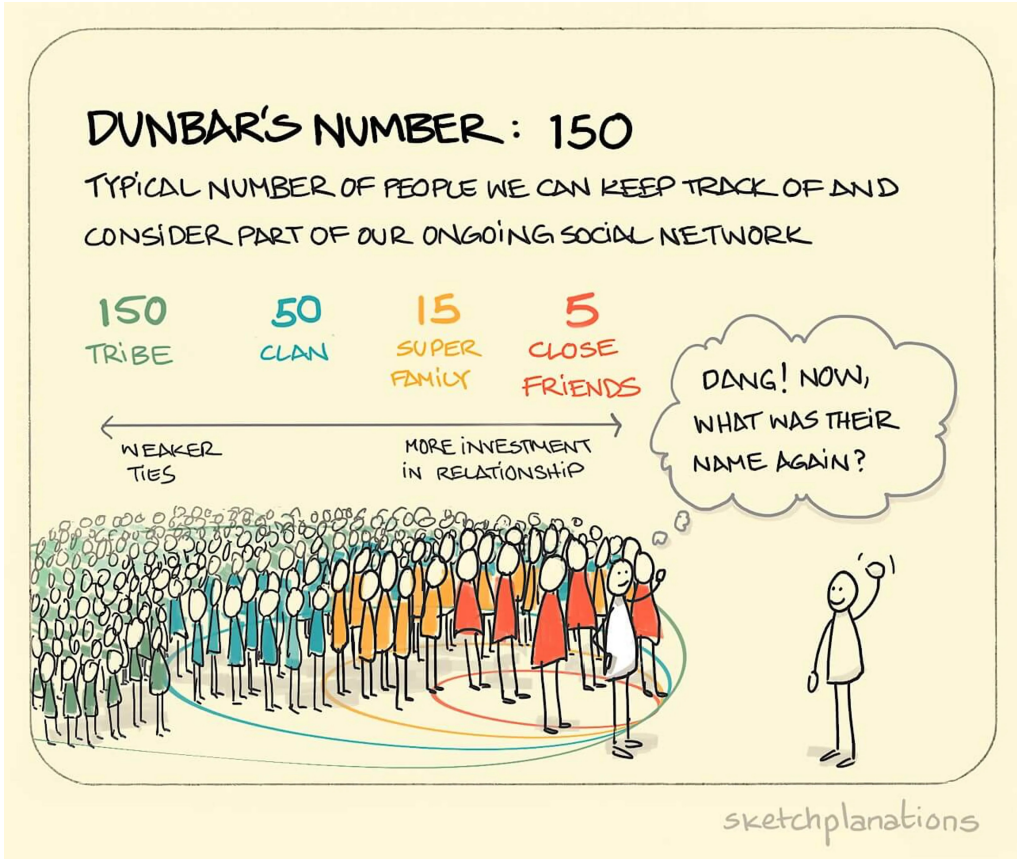
[1]Harvard University, Cambridge, MA, USA. [2]Massachusetts Institute of Technology, Cambridge, MA, USA. [3]University of Michigan, Ann Arbor, MI, USA. [4]New York University, New York, NY, USA. [5]Northeastern University, Boston, MA, USA. [6]Interdisciplinary Scientific Research, Seattle, WA, USA. [7]Northwestern University, Evanston, IL, USA.

Example: Dunbar's number

- How big is a "typical" social network?
- By analyzing social networks in primates, Dunbar hypothesized that there are cognitive limits to the size of one's social network:
  - *The limit imposed by neocortical processing capacity is simply on the number of individuals with whom a stable inter-personal relationship can be maintained*
- Dunbar estimated that the typical size of a human social network is 150 individuals

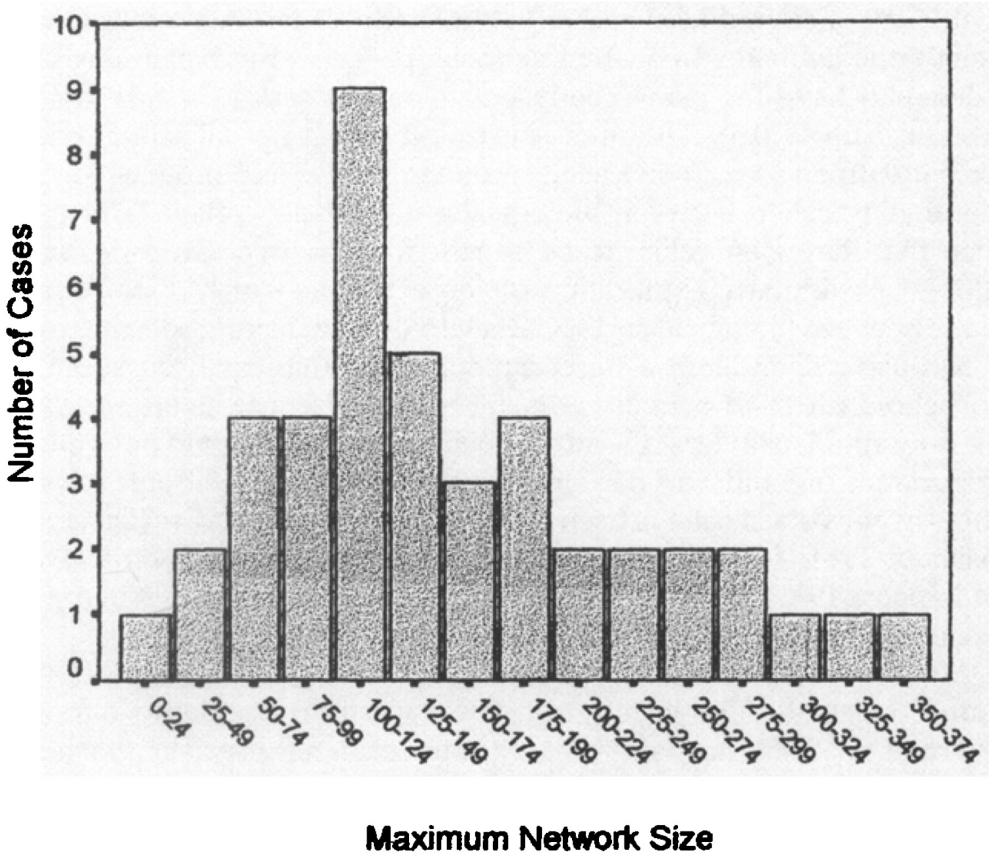
Dunbar's number

- Exchange of Christmas cards in a population of 43 families (surveys)
- Trajectories data of 2,984 cards



Dunbar's number

- The average number of cards sent is 68.19 (range 11-149)
- Since many cards were sent by couples/families, the average size of the network is 153.5 (+-84.5)



Dunbar's number on Twitter

- In Gonçalves, Perra, & Vespignani, 2011 the authors validated Dunbar's number for Twitter

Dunbar's number on Twitter

- In Gonçalves, Perra, & Vespignani, 2011 the authors validated Dunbar's number for Twitter
- *We analyze a dataset of Twitter conversations collected across six months involving* ***1.7 million individuals*** *and test the theoretical cognitive limit on the number of stable social relationships known as Dunbar's number. We find that* ***the data are in agreement with Dunbar's result****; users can entertain a maximum of 100–200 stable relationships. Thus, the 'economy of attention' is limited in the online world by cognitive and biological constraints as predicted by Dunbar's theory.*

Dunbar's number on Twitter

- *On one hand, it can be argued that microblogging tools facilitate the way we handle social interactions and that this results in an online world where human social limits are finally lifted, making predictions such as the Dunbar's number obsolete.*

Dunbar's number on Twitter

- *On one hand, it can be argued that microblogging tools facilitate the way we handle social interactions and that this results in an online world where human social limits are finally lifted, making predictions such as the Dunbar's number obsolete.*
- *Microblogging and online tools, on the other hand, might be analogous to a pocket calculator that, while speeding up the way we can do simple math, does not improve our cognitive capabilities for mathematics.*

Information Networks



Information Networks

- Information online has a fundamental network structure
- Example: Links among web pages can help us understand
- How these pages are related
- How they are grouped in different communities
- Which pages are the most important

Information Networks: Divided they Blog

- In 2005 Adamic and Glance measured the degree of interaction between liberal and conservative blogs
- They wanted to uncover any differences in the structure of the two communities
- They analyzed the posts of 1,494 blogs over the period of 2 months before the 2004 Presidential Election
- They studied how often they referred to one another and quantify the overlap in the topics they discussed (intra and inter communities)
- They found differences in the behavior of liberal and conservative blogs
  - Conservative blogs link to each other more frequently and in a denser pattern

Information Networks: Divided they Blog

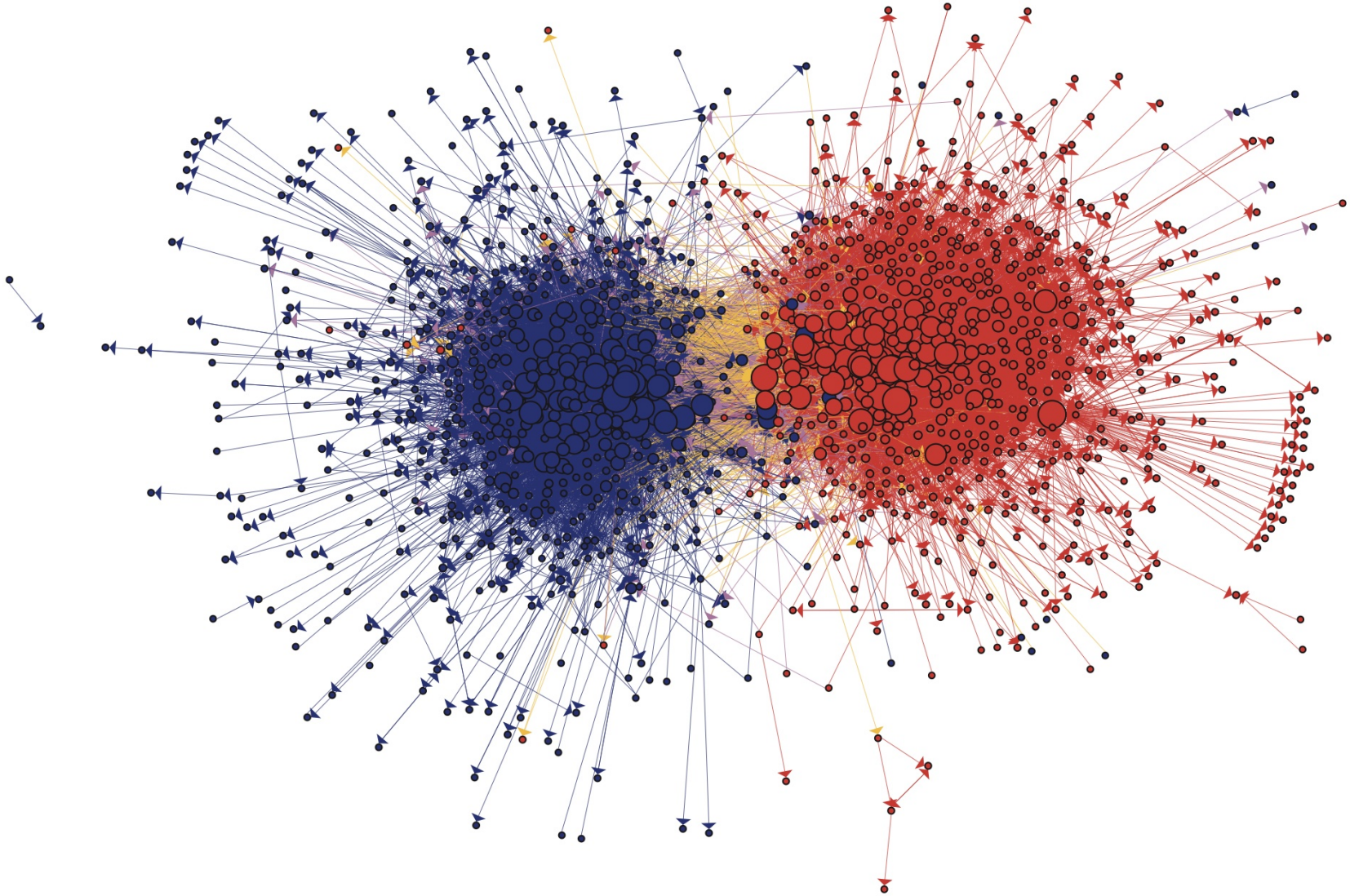
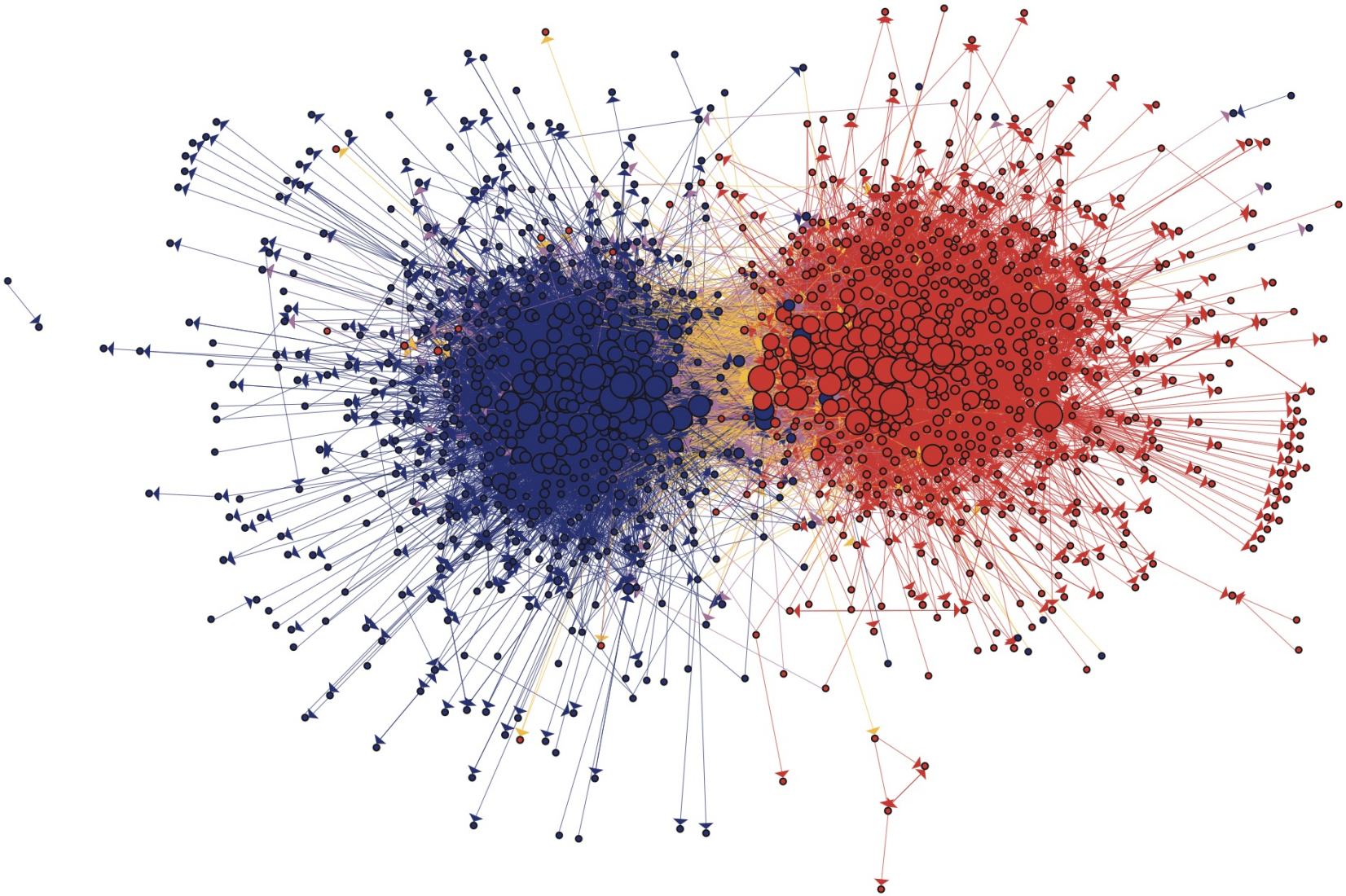


Image from Adamic and Glance, The Political Blogosphere and the 2004 U.S. Election: Divided They Blog, ACM 2005

Information Networks: Divided they Blog

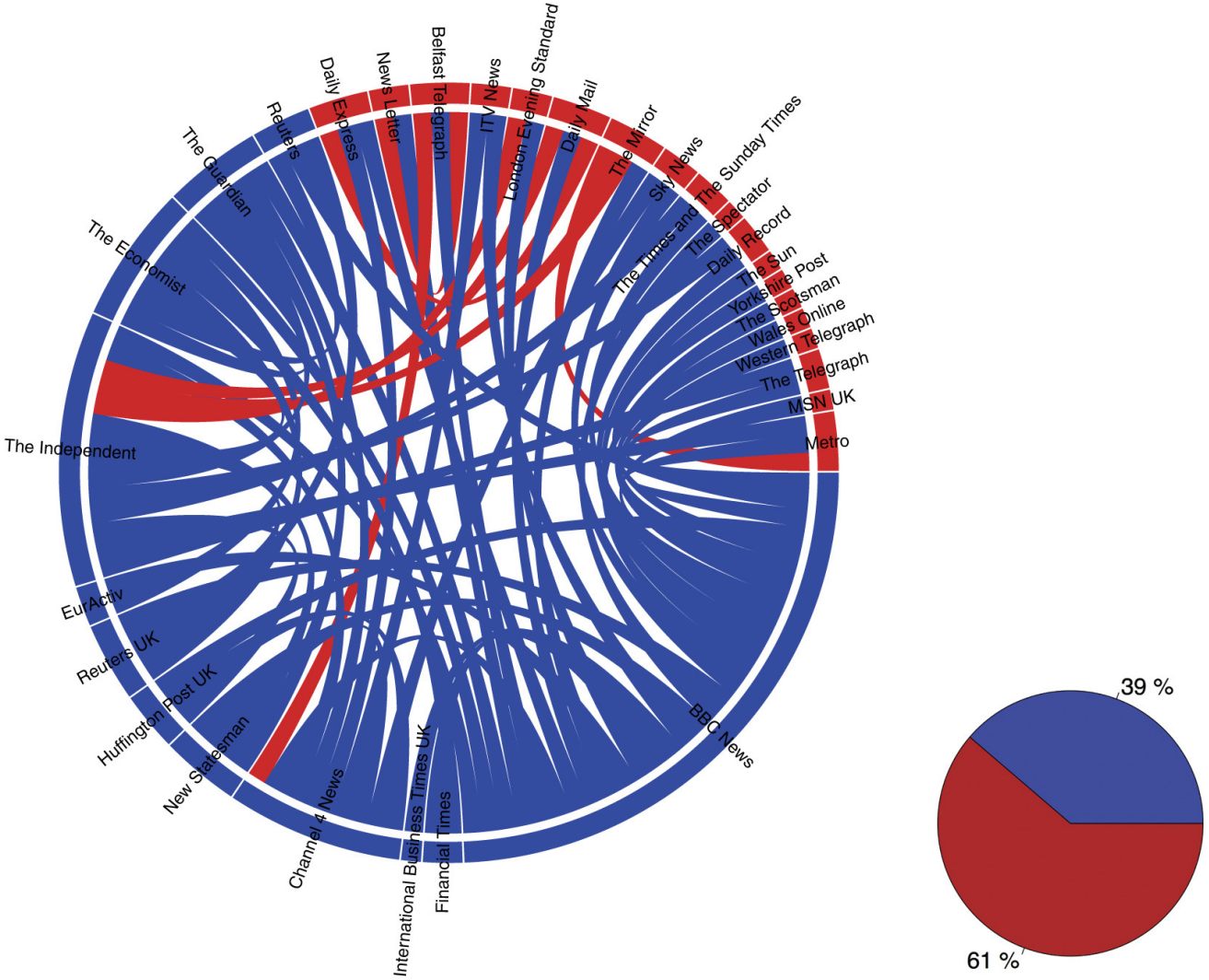
The network structure reveals two natural and well-separated clusters



Information Networks: The Brexit Debate on Facebook

- UK news sources on Facebook
- We filtered the posts related to Brexit in the period Jan-Jul 2016
- We built a network involving more than 1M users
- Links among pages depend on users' interaction with content (likes)

Information Networks: The Brexit Debate on Facebook



(a)

(b)

Image from Del Vicario, Zollo, et al., Mapping social dynamics on Facebook: The Brexit debate. Social Networks, Vol. 50, 2017